

# The Bolder the People, the Greater the Harvest: Ecological Degradation and the Great Leap Forward Famine

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## Abstract

This paper provides a critical analysis of the catastrophic humanitarian consequences of Maoist agricultural policies in the People's Republic of China between 1959 and 1961, more often known as the "Great Leap Forward." Existing literature treats the famine primarily as a political or institutional failure, but this paper explores more deeply the factors of ecological degradation. Specifically, We argue that indeed Maoist agricultural policies distorted incentives and misallocated resources in Mainland China. But more importantly, these policies systematically weakened the ecological foundation of agricultural production through soil organic matter depletion, deforestation, erosion, and the disruption of traditional land-management practices. Using a Bayesian latent-state production model fitted to national aggregate data from 1952 to 1977, we show that the timing of policy-driven ecological vulnerability corresponds closely to the collapse in grain output. Counterfactual simulations suggest that the removal of policy stress alone would have raised annual grain output by approximately 43.3 million tons during the famine years (1959–1961). The case of Sichuan Province further illustrates how political exaggeration, extraction pressure, and ecological degradation interacted to produce severe famine outcomes.

**Keywords:** Mao Zedong; Great Leap Forward; Ecological Degradation; Bayesian Latent Variable Model

# 1 Introduction

The millions of excess deaths and lost births during China’s Great Famine of 1959–1961 constituted one of the most devastating demographic catastrophes of the twentieth century. Chinese government and academic circles have advanced competing explanations for the atrocities of the Great Leap Forward. The Communist Party of China’s post-Mao canonical formulation appears in the 1981 *Resolution on Certain Questions in the History of Our Party Since the Founding of the PRC*. In the section dealing with 1959–1961, the Resolution states that the country suffered “serious difficulties” and “heavy losses,” caused mainly by the errors of the Great Leap Forward and the anti-right deviation campaign, together with natural disasters and the Soviet government’s tearing up of contracts [CPC \(1981\)](#).

By contrast, academic scholarship has emphasized different priorities while, to varying degrees, accepting aspects of the CPC’s canonical narrative. The first group emphasizes natural shocks, arguing that droughts, floods, extreme rainfall, or other climatic anomalies were the main cause of agricultural decline and the food crisis. The second group emphasizes institutional and policy factors, arguing that agricultural crises are often not simply the result of nature, but are instead due to procurement policies, forced collectivization, imbalances in labor allocation, radical production campaigns, and unreasonable land-use systems that weakened the resilience of the agricultural system.

Regarding the disasters during the Great Leap Forward, [Houser et al. \(2009\)](#) constructed a Bayesian latent index model using provincial-level mortality panels, attempting to distinguish nationwide policy shocks from regional weather shocks, and concluded that national policies were overall more important than weather. Meanwhile, [Li and Yang \(2005\)](#), starting from food output itself and using a provincial-level production function approach, pointed out that resource transfers and excessive procurement were the main sources of output collapse during the Great Leap Forward.

The starting point of this paper is that if one only observes the ultimate population outcomes, it is difficult to distinguish which losses came from short-term weather anomalies and which from the long-term destruction of the agricultural production system by institutional factors. Agricultural output and soil conditions, situated between policy shocks and population outcomes, constitute an intermediate layer that is better suited to identifying how institutions shape agricultural vulnerability. This paper therefore attempts to combine [Houser et al. \(2009\)](#)’s latent policy index identification method with [Li and Yang \(2005\)](#)’s agricultural production function approach, explicitly modeling “soil degradation / ecological vulnerability” as a latent state variable, to explain why erroneous policies not only directly reduce output but also make the agricultural system more susceptible to natural disasters.

# 2 Background

Following its victory in the Chinese Civil War (1945–1949), the Communist Party of China, under the leadership of Mao Zedong, consolidated its political control over Mainland China. Yet, after eight years of the Second World War against Japan and four years of Civil War, the nation became heavily devastated and largely isolated from the international community. The Communist Party of China therefore faced

the urgent task of consolidating its political authority, restructuring both urban and rural society, and strengthening relations with the Soviet Union.

In the early 1950s, the consolidation of the new regime and the widely proclaimed achievements of the First Five-Year Plan fostered growing confidence among Chinese Communist leaders in the viability of rapid state-led transformation [Britannica \(n.d.-a,n\)](#). The march to socialism seemed to go along reasonably well with the dictates of industrial development, and the determination and fundamental optimism of the Communist leaders appeared justified, especially in view of the decades of invasion, disintegration, self-doubt, and humiliation that had shaped China before 1949 [Britannica \(n.d.-c\)](#). This atmosphere of growing confidence also resonated with Mao's broader strategic outlook. As early as August 1946, during his talk with the American correspondent Anna Louise Strong, Mao had argued that "all reactionaries are paper tigers" [Mao \(1946\)](#), reflecting his enduring conviction that seemingly powerful obstacles could ultimately be overcome through political will and revolutionary struggle.

By the mid-1950s, the confidence generated by early state-building successes began to converge with Mao's longer-standing revolutionary optimism, helping to shape the increasingly radical ambitions he would articulate in 1957. On November 6, 1957, thousands of delegates from sixty-four communist and workers' parties around the world gathered in the brand-new Central Lenin Stadium (now Luzhniki Stadium) in Moscow to celebrate the fortieth anniversary of the October Revolution. In his long keynote speech, Soviet leader Nikita Khrushchev boasted that the Soviet Union would catch up to or even surpass the United States in fifteen years in terms of total output of major industrial products. At a follow-up meeting on November 18, Chairman Mao Zedong, inspired by Khrushchev's speech, made an impromptu declaration that China would follow the example of the Soviet Union and surpass Britain in fifteen years. What Mao had in mind was to surpass Britain mainly in terms of steel output, which he considered to be synonymous with industrialization [Zhu \(2023\)](#).

Later, this radical developmental vision was institutionalized at the Second Session of the Eighth Party Congress, held from May 5 to May 23, 1958, which formally endorsed the General Line of Socialist Construction and called on China to catch up with and surpass Britain in major industrial products within fifteen years or even less. Mao Zedong spoke at the meeting, emphasizing the need to break with blind faith, liberate thought, and foster a creative spirit that dared to think, speak, and act [CPC News \(1958\)](#). After the meeting, a nationwide upsurge of the Great Leap Forward rapidly took shape.

## 2.1 Environmental Effects of Great Leap Policies

To understand why ecological degradation should be treated as a central explanatory mechanism rather than a secondary consequence, it is necessary to examine the specific policy environment of the Great Leap Forward. The environmental damage of the late 1950s did not emerge spontaneously. It was produced through a set of highly politicized campaigns that systematically pushed agricultural practice beyond local ecological limits.

**Reorganization of rural production.** The people's commune system reorganized rural production from the ground up. Communalization gutted household incentives and handed farming decisions to cadres who answered to political targets rather than to what the land could actually sustain. Soil conditions, seasonal rhythms, local ecological limits — none of it counted the way Party directives did. Guangdong had already shown where this was heading. Fang Fang, a senior provincial leader who pressed for gradual, locally calibrated land reform, became the principal target of a 1952 campaign against "right deviation" and "localism" after Mao himself oversaw the discussions on Guangdong's rural policy. [Song \(2007\)](#) The message was that radical campaigns backed from Beijing called the shots; nobody was adjusting policy to what was happening on the ground. The agricultural system lost its capacity for self-correction.

**Pseudoscientific agricultural ideas.** Another important factor was the spread of pseudoscientific agricultural ideas. As aforementioned, Fang Fang's wrongful persecution was not outright resistance to Mao's policies, but his preference for a more gradual and locally grounded approach to rural reform. On August 27, 1958, the *People's Daily* published an editorial entitled "The Bolder the People Are, the Greater the Harvest Will Be" [People's Daily \(1958a\)](#). In the article, the author claimed that past beliefs that dry crops required minimal water and that excessive fertilizer harmed the soil were wrong. The new approach in Shouzhong emphasized that sufficient water, fertilizer (up to 1,000 kg per mu or 667 m<sup>2</sup>), and deep plowing (about 1-1.5 meter) were essential for high yields. This method enabled dense planting, such as millet grown with up to 260,000 plants per mu without reducing grain size.

These pseudoscientific ideas led to another major source of ecological damage: irrational cultivation practices promoted under production pressure. In the campaign atmosphere of the Great Leap Forward, local cadres were encouraged to pursue unrealistic yield targets and to demonstrate political enthusiasm through visibly radical techniques. Among the most famous examples were close planting and deep plowing. In theory, these methods were presented as scientific breakthroughs that could dramatically increase output. In practice, crop competition was intensified: for light, water, and nutrients, while deep tillage disrupted soil structure and reduced aeration in the long term. Rather than improving productivity, ironically, these practices accelerated soil exhaustion and undermined future growing capacity. In ecological terms, the agricultural system began to consume its own productive foundation.

**Land expansion and deforestation.** A third channel operated through land expansion and deforestation. The simultaneous drive for backyard steel production and increased cultivated acreage encouraged the large-scale cutting of trees and the opening of marginal land. Forest clearance reduced vegetation cover, increased topsoil erosion, lowered water retention, and weakened local microclimatic stability. Bringing fragile or unsuitable land under cultivation may have temporarily expanded sown area on paper, but it also made agricultural production more sensitive to drought and flooding. Thus, policies aimed at rapid industrial and agricultural expansion likely reduced the resilience of the land itself.

**The Four Pests Campaign.** Another striking example was the Four Pests Campaign, especially the campaign against sparrows. This policy illustrates how state-led ecological intervention could generate severe unintended consequences. By disrupting the food chain and reducing natural predation, the elimination of sparrows contributed to surges in insect populations, which in turn damaged crops. The significance of this case lies not only in its dramatic symbolism, but also in what it reveals about the logic of the period: environmental systems were treated as politically malleable and scientifically transparent, even when intervention rested on deeply flawed assumptions. In this sense, ecological imbalance was not accidental. It was built into the policy style of the Great Leap Forward.

These policies matter for this paper because they provide the historical basis for treating soil degradation and ecological vulnerability as analytically central. The argument here is not that every local decline in output can be directly measured through a separate soil indicator. Rather, it is that the political campaigns of the late 1950s created strong reasons to expect cumulative environmental damage, and that such damage would have made some regions more vulnerable than others when adverse weather shocks arrived. This is precisely why the empirical sections of this paper use data primarily as supporting evidence rather than as the sole basis of the argument. The historical record already indicates that policy distorted not only incentives and procurement, but also the ecological conditions under which agricultural production took place.

For this reason, the Great Leap Forward should not be interpreted simply as a case of bad weather coinciding with procurement excess. It should be understood as a moment when radical political ambition reshaped rural institutions, transformed cultivation practices, and weakened the ecological resilience of agriculture. The environmental dimension is therefore not a side issue, but one of the key channels through which policy failure was converted into agricultural collapse.

### 3 Literature Review

[Houser et al. \(2009\)](#) use a Bayesian panel analysis of provincial mortality rates to distinguish nationwide policy shocks from local weather shocks during China's Great Famine. They find that about 70% of excess deaths during 1959–1961 can be attributed to national policy effects rather than weather alone, supporting the view that the famine was predominantly man-made. However, because their analysis relies on mortality outcomes and does not directly observe grain production, grain movements, or ecological conditions, it cannot fully identify the internal mechanisms through which agricultural systems translated policy shocks into famine outcomes.

[Li and Yang \(2005\)](#) move closer to this production-side mechanism. Their theoretical and empirical analysis shows that central planners overestimated the productivity gains from collectivization, which led to excessive diversion of agricultural resources into industry and an overly aggressive grain procurement policy. These policies reduced rural caloric intake, weakened labor productivity, and contributed to the collapse of grain output; their panel estimates suggest that resource diversion and excessive procurement explain about 61% of the decline in grain output.

However, distinguishing only between policy and weather shocks is still insufficient for understanding the agro-ecological dynamics of famine. Agricultural production is not a static system that responds mechanically to external shocks; it is

embedded in soil fertility, water regimes, cropping structures, and land-management institutions. Soil and land degradation are cumulative processes. Unsustainable land use, nutrient depletion, erosion, and the loss of soil organic matter can gradually reduce biological productivity and weaken the resilience of agricultural systems [FAO and ITPS \(2015\)](#); [Lal \(2004\)](#). Climate shocks such as droughts, excessive rainfall, and abnormal humidity may therefore have stronger effects in regions where prior policies had already weakened ecological conditions [IPCC \(2019\)](#).

This paper extends the existing literature by treating ecological degradation as a latent, path-dependent state variable rather than as an unexplained residual. In this framework, policy shocks may increase famine severity not only through direct channels such as procurement and labor diversion, but also through ecological channels that make soil and local production systems more sensitive to weather and moisture shocks. Empirically, mortality and agricultural output are used as outcome variables, while regional environmental conditions, historical land-use pressure, and policy implementation intensity help identify how ecological degradation mediated the relationship between policy, weather, and famine severity.

## 4 Theoretical Framework and Hypotheses

This paper proposes a three-layer mechanism linking Maoist agricultural policies, ecological degradation, and agricultural collapse during the Great Leap Forward.

**First Factor: Direct output reduction.** Radical policy interventions directly reduced agricultural output. Excessive procurement, the diversion of rural labor to steel production and large-scale nonagricultural projects, the weakening of household incentives under collectivized organization, and unrealistic production targets all reduced effective agricultural input and distorted local production decisions.

**Second Factor: Ecological foundation damage.** These policies also altered the ecological foundation of farming. High-intensity cultivation, reduced fallowing, weakened land maintenance, deforestation, and the neglect of soil conservation likely accelerated the depletion of soil organic matter, reduced water retention, and increased erosion risk. Even when these processes did not immediately produce a visible catastrophe, they increased the long-run fragility of the agricultural system.

**Third Factor: Amplification of weather shocks.** Once ecological resilience had been weakened, the same degree of weather fluctuation could generate much larger output losses, also indicated by [Houser et al. \(2009\)](#). In this sense, ecological degradation serve as an amplifier of exogenous shocks. The agricultural crisis of the Great Leap Forward should therefore be understood neither as a simple weather disaster, nor solely as a procurement crisis, but as the result of an interaction between political misallocation, declining food availability, and the cumulative erosion of ecological resilience.

Based on this mechanism, this paper proposes three hypotheses:

**H1:** National-level policy shocks significantly reduced agricultural output.

**H2:** Policy shocks exerted an indirect negative effect on output through the accumulation of soil degradation and ecological vulnerability.



**H3:** Policy and weather interacted significantly, such that politically induced ecological fragility amplified the damage caused by adverse weather shocks.

## 5 Data and Mathematical Model

This section uses national aggregate agricultural data to test whether the timing of Great Leap Forward policy pressure is consistent with a temporary rise in ecological vulnerability and a collapse in grain output. Because direct annual measurements of soil organic matter are not available for the Great Leap Forward period, this paper does not claim to measure soil organic matter loss directly. Instead, it models ecological degradation as a latent vulnerability state. This latent state captures the cumulative pressure placed on the agricultural system through excessive procurement, labor diversion, land-use intensity, and radical farming campaigns.

### 5.1 Data

The national data include annual observations from 1952 to 1977 on grain output, grain procurement, retained grain per capita, rural labor, grain-sown area, draft animals, farm machinery, and chemical fertilizer. These variables allow the model to connect three layers: observed agricultural inputs, policy stress, and unobserved ecological vulnerability. The study by [Li and Yang \(2005\)](#) is especially useful here because it shows that the Great Leap Forward production collapse was closely related to resource diversion and excessive procurement; their analysis attributes 61 percent of the decline in grain output to these two policies. Table 1 presents the full dataset.

### 5.2 Model Specification

Let  $Y_t$  denote national grain output in year  $t$ . Let  $X_t$  represent observed production inputs, including rural labor, grain-sown area, draft animals, machinery power, and chemical fertilizer. The model defines a policy-stress vector:

$$P_t = (\text{GLF}_t, E_t, L_t, I_t) \quad (1)$$

where  $\text{GLF}_t$  is a dummy variable equal to one for 1958–1961;  $E_t$  is excess procurement above the 1952–1957 average procurement rate;  $L_t$  measures labor diversion relative to 1957; and  $I_t$  measures land-use intensity, defined as grain-sown area per rural worker relative to 1957. The latent ecological vulnerability state is modeled as:

$$S_t = \rho S_{t-1} + \omega' P_t \quad (2)$$

where  $0 < \rho < 1$  measures persistence, and  $\omega > 0$  means that policy stress increases ecological vulnerability. The grain output equation is:

$$\log(Y_t) = \alpha + \sum_j \beta_j \log(X_{jt}) - \delta S_t + \epsilon_t \quad (3)$$

with  $\epsilon_t \sim N(0, \sigma^2)$ . The key parameter is  $\delta$ . If  $\delta > 0$ , then higher ecological vulnerability is associated with lower grain output. This fits the central argument of this paper: Maoist agricultural policy did not only reduce output through procurement and labor diversion; it also weakened the agricultural system’s capacity to absorb stress.

Table 1: National Agricultural Data, 1952–1977

| Year | Grain<br>Output<br>(Mt) | Grain<br>Proc.<br>(Mt) | Ret. per<br>Capita<br>(kg) | Rural<br>Labor<br>(M) | Area<br>Sown<br>(M ha) | Draft<br>Anim.<br>(M Head) | Mach.<br>(M HP) | Fert.<br>(Mt) |
|------|-------------------------|------------------------|----------------------------|-----------------------|------------------------|----------------------------|-----------------|---------------|
| 1952 | 164                     | 33                     | 260                        | 173                   | 124                    | 76                         | 0.3             | 0.08          |
| 1953 | 167                     | 47                     | 242                        | 177                   | 127                    | 81                         | 0.4             | 0.12          |
| 1954 | 170                     | 51                     | 228                        | 182                   | 129                    | 85                         | 0.5             | 0.16          |
| 1955 | 184                     | 48                     | 256                        | 186                   | 130                    | 88                         | 0.8             | 0.24          |
| 1956 | 193                     | 40                     | 284                        | 185                   | 136                    | 88                         | 1.1             | 0.33          |
| 1957 | 195                     | 46                     | 273                        | 193                   | 134                    | 84                         | 1.7             | 0.37          |
| 1958 | 200                     | 52                     | 268                        | 155                   | 128                    | 78                         | 2.4             | 0.55          |
| 1959 | 170                     | 64                     | 193                        | 163                   | 116                    | 79                         | 3.4             | 0.54          |
| 1960 | 143                     | 47                     | 182                        | 170                   | 122                    | 73                         | 5.0             | 0.66          |
| 1961 | 148                     | 37                     | 209                        | 197                   | 121                    | 69                         | 7.1             | 0.45          |
| 1962 | 160                     | 32                     | 229                        | 213                   | 122                    | 70                         | 10              | 0.63          |
| 1963 | 170                     | 37                     | 231                        | 220                   | 121                    | 75                         | 12              | 1.0           |
| 1964 | 188                     | 40                     | 256                        | 228                   | 122                    | 79                         | 13              | 1.3           |
| 1965 | 195                     | 39                     | 261                        | 234                   | 120                    | 84                         | 15              | 1.9           |
| 1966 | 214                     | 41                     | 282                        | 243                   | 121                    | 87                         | 17              | 2.3           |
| 1967 | 218                     | 41                     | 281                        | 252                   | 119                    | 90                         | 20              | 2.4           |
| 1968 | 209                     | 40                     | 261                        | 261                   | 116                    | 92                         | 22              | 2.7           |
| 1969 | 211                     | 38                     | 259                        | 271                   | 118                    | 92                         | 26              | 3.1           |
| 1970 | 240                     | 46                     | 282                        | 278                   | 119                    | 94                         | 29              | 3.4           |
| 1971 | 250                     | 44                     | 293                        | 284                   | 121                    | 95                         | 38              | 3.8           |
| 1972 | 241                     | 39                     | 298                        | 283                   | 121                    | 96                         | 50              | 4.3           |
| 1973 | 265                     | 48                     | 293                        | 289                   | 121                    | 97                         | 65              | 4.8           |
| 1974 | 275                     | 47                     | 303                        | 292                   | 121                    | 98                         | 81              | 5.4           |
| 1975 | 285                     | 53                     | 304                        | 295                   | 121                    | 97                         | 102             | 6.0           |
| 1976 | 286                     | 49                     | 306                        | 294                   | 121                    | 95                         | 117             | 6.8           |
| 1977 | 283                     | 48                     | 300                        | 293                   | 120                    | 94                         | 140             | 7.6           |

*Note:* Mt = million tons; M ha = million hectares; M HP = million horsepower; Proc. = procurement; Anim. = animals; Ret. = retained; Mach. = machinery; Fert. = fertilizer. Data compiled from national agricultural statistics.



### 5.3 Prior Specification and Estimation

The model uses Bayesian priors to prevent overfitting, which is necessary because the national time series contains only twenty-six annual observations. Positive priors are placed on production-input elasticities, policy-stress loadings, and the ecological-damage penalty. The persistence parameter  $\rho$  is given a prior centered on a relatively durable but not permanent soil-degradation process. In other words, ecological vulnerability can rise rapidly during a policy shock, but it does not disappear immediately after the policy shock ends. Estimation is performed via MAP optimization (BFGS) followed by a Laplace approximation for posterior inference.

## 6 Empirical Results

### 6.1 National Descriptive Evidence

The national data strongly support the timing of the paper’s argument. Grain output averaged 190.7 million tons during 1955–1957, but only 153.7 million tons during 1959–1961, a decline of 19.4%. The lowest output occurred in 1960, when national grain production fell to 143 million tons. At the same time, procurement pressure rose sharply. In 1957, grain procurement was 46 million tons, but in 1959 it reached 64 million tons, even though grain output had already fallen. Retained grain per capita fell from 273 kg/person in 1957 to 193 kg/person in 1959 and 182 kg/person in 1960. These figures show that the crisis was not only a production shock; it was also a distribution and extraction shock.

### 6.2 Bayesian Latent-State Estimates

The Bayesian model estimates a strong persistence effect in the latent ecological-vulnerability state. The posterior mean of the soil-persistence parameter is:

$$\rho = 0.939 \tag{4}$$

This implies that ecological stress did not disappear immediately after policy pressure was relaxed. The estimated latent vulnerability index rises sharply from  $-0.87$  in 1957 to  $1.21$  in 1959, peaks at  $2.05$  in 1960, and remains high at  $1.92$  in 1961, as shown in Figure 1.

Table 2: Latent Ecological Vulnerability Index and Observed Grain Output

| Year | Observed Grain Output (Mt) | Latent Vulnerability Index | 90% Interval     |
|------|----------------------------|----------------------------|------------------|
| 1957 | 195                        | $-0.87$                    | $[-1.23, -0.51]$ |
| 1958 | 200                        | $-0.18$                    | $[-0.51, 0.15]$  |
| 1959 | 170                        | $1.21$                     | $[0.87, 1.55]$   |
| 1960 | 143                        | $2.05$                     | $[1.66, 2.44]$   |
| 1961 | 148                        | $1.92$                     | $[1.52, 2.32]$   |
| 1962 | 160                        | $1.61$                     | $[1.21, 2.01]$   |

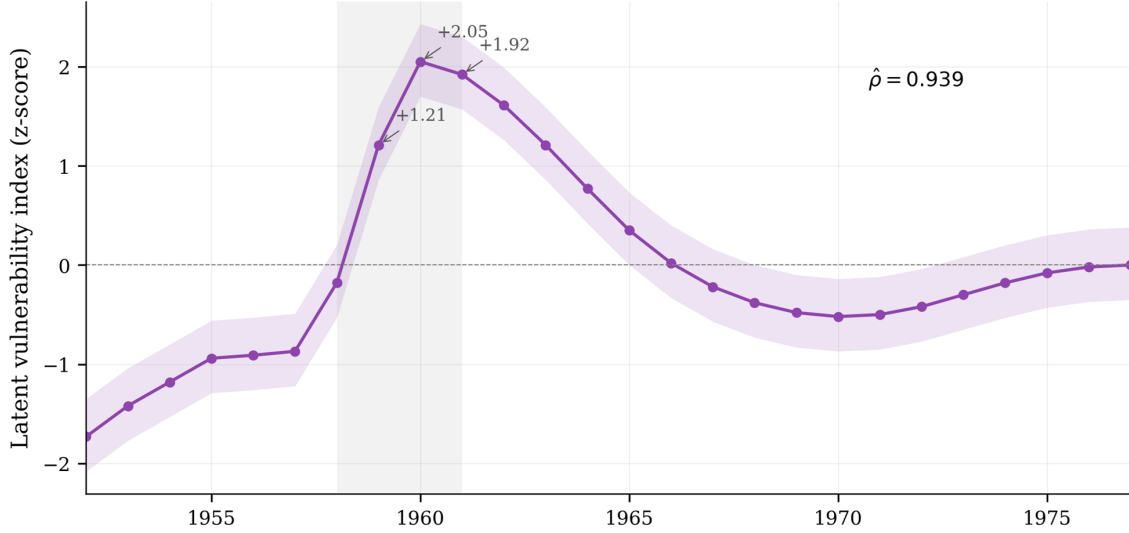


Figure 1: Latent ecological vulnerability index, 1952–1977. Shaded region: Great Leap Forward period (1958–1961). Ribbon: 90% credible interval. The posterior mean of the persistence parameter is  $\hat{\rho} = 0.939$ .

### 6.3 Counterfactual Simulation

The counterfactual simulation removes Great Leap Forward policy stress from 1958–1961 while keeping observed national production inputs unchanged. Under this scenario, the model estimates that national grain output would have been substantially higher during the worst famine years. The posterior average policy-related output

Table 3: Counterfactual Policy-Removal Scenario

| Year | Observed Output (Mt) | Counterfactual (Mt) | Policy Gap (Mt) |
|------|----------------------|---------------------|-----------------|
| 1959 | 170.0                | 195.4               | 25.4            |
| 1960 | 143.0                | 204.1               | 61.1            |
| 1961 | 148.0                | 201.4               | 53.4            |

gap for 1959–1961 is:

$$\text{Gap} = 43.3 \text{ million tons per year} \quad (5)$$

with a 90% interval of [28.3, 55.9], as shown in Figure 2.

This result should not be interpreted as a precise measurement of lost soil organic matter. The data are not sufficient. Rather, the result should be interpreted as evidence that the national agricultural system behaved as if its ecological resilience had been weakened. The timing of the modeled vulnerability index corresponds closely to the historical evidence discussed earlier: deep plowing, close planting, deforestation, labor diversion, communal organization, and pseudoscientific production campaigns all created conditions under which soil structure, water retention, and local ecological balance could deteriorate.

This finding also fits the institutional interpretation of the famine. [Meng et al. \(2010\)](#) argue that the famine cannot be explained by aggregate food shortage alone.

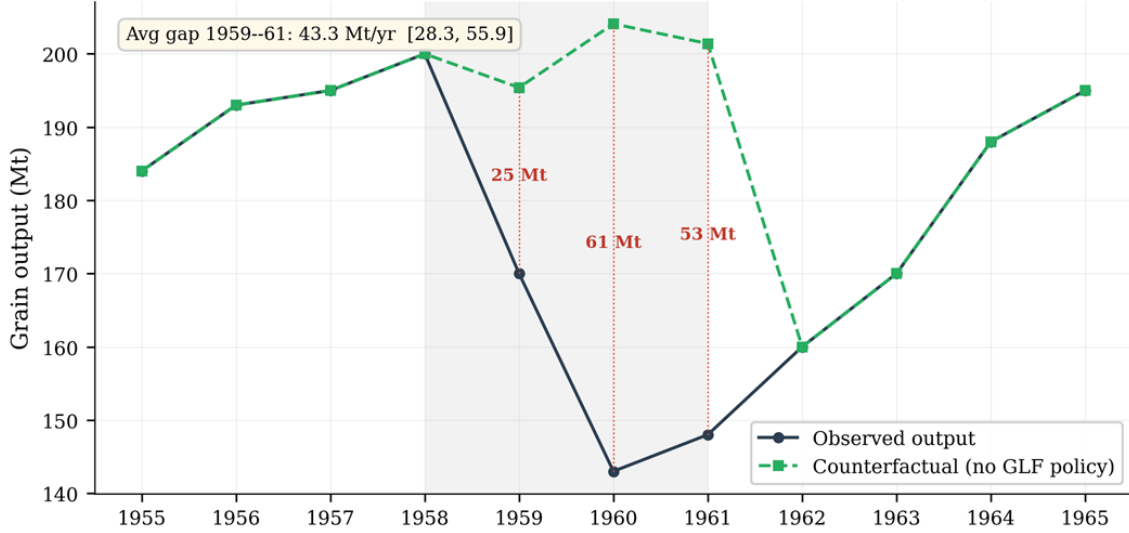


Figure 2: Counterfactual simulation: observed vs. no-GLF-policy grain output, 1955–1965. The posterior average output gap for 1959–1961 is 43.3 million tons per year (90% credible interval: [28.3, 55.9]).

Their NBER working paper reports that in 1959, when the famine began, food production was almost three times the level required for subsistence, and that regions with higher per capita food production suffered higher mortality—a reversal of the normal relationship between production and famine risk. This supports the interpretation that procurement, distorted information, and political allocation mechanisms converted a production shock into a mass famine.

## 7 Case Study of Sichuan Province

Throughout the Chinese history, Sichuan Province is famously known for its name of 'Land of Abundance', thanks to its fertile soil, moist and warm environment. However, demographic evidence from [Peng \(1987\)](#) documents a crude death rate of approximately 47 per thousand in Sichuan in 1959, and existing literature, like [Houser et al. \(2009\)](#) haven't provide a reasonable explanation for this scenario. Currently, our model above uses national annual data, this paper does not estimate a true Sichuan provincial production function. A full Sichuan model would require county-level data on grain output, procurement, mortality, labor, weather, land-use intensity, and soil conditions. Nevertheless, Sichuan is essential to the argument because it illustrates how national policy mechanisms operated in a high-intensity provincial case.

Sichuan was not a marginal agricultural region. It was historically known as a major grain-producing province as aforementioned, and during the Great Leap Forward it was publicly represented as a site of extraordinary agricultural success. This makes its later famine severity especially important. If a province officially described as abundant became one of the worst famine zones, then the cause cannot be reduced to simple natural scarcity. Instead, Sichuan shows how false reporting, procurement pressure, and local implementation could turn claimed abundance into rural starvation.

## 7.1 Propaganda

The *People's Daily* provides crucial evidence of the political information environment. On October 15, 1958, the *People's Daily* reported that Sichuan's grain and cotton output had both increased by more than 100%. The article claimed that Sichuan's total grain output would reach 96 billion jin (48 billion kg), compared with 46.5 billion jin (23.5 billion kg) the previous year, and that average grain yield would reach 1,020 jin per mu (7.65 t/ha). The report also described Sichuan as one of China's major surplus-grain provinces and stated that the "great harvest" would allow the province to make a larger contribution to national construction [People's Daily \(1958b\)](#).

A few days later, on October 21, 1958, the *People's Daily* intensified this narrative by declaring that Sichuan had become a "grain thousand-jin province." The article claimed that Sichuan's 70 million people would have an average of 1,333 jin (666.5 kg) of grain per person. It also stated that more than 10 million people had been organized into field "battle corps" to carry out emergency harvesting, while nearly half of the grain and cotton still remained in the fields [People's Daily \(1958c\)](#).

These reports should not be used as reliable evidence of actual output, but could be used as evidence of political exaggeration. The language of "battle corps," "thousand-jin province," and "great harvest" shows how agricultural production was represented through military and ideological categories. This propaganda environment reported output influenced procurement expectations, which become dangerously detached from actual rural food availability.

## 7.2 Propaganda Model

This mechanism can be formalized as:

$$\text{ReportedOutput}_{ct} = \text{ActualOutput}_{ct} + \text{Exaggeration}_{ct} \quad (6)$$

$$\text{ProcurementPressure}_{ct} = \frac{\text{StateProcurement}_{ct}}{\text{ActualOutput}_{ct}} \quad (7)$$

where  $c$  refers to county and  $t$  refers to year. If officials set procurement based on reported output rather than actual output, then an increase in  $\text{Exaggeration}_{ct}$  raises the risk that procurement will exceed what rural communities can survive. In this way, false abundance could produce real scarcity.

## 7.3 Mortality Evidence

Sichuan's mortality record supports this interpretation. [Peng \(1987\)](#) reports that in 1959 the crude death rate reached about 47 per thousand in Sichuan, compared with 31 per thousand in Guangxi and 20 per thousand in Guizhou. This extreme mortality is difficult to reconcile with the *People's Daily* image of Sichuan as a province of extraordinary grain abundance. This contradiction suggests that the famine was exacerbated by the interaction between exaggerated reporting, extraction, and local political pressure.

[Bramall's \(2011\)](#) county-level study of Sichuan is especially important here. Using county-level data on mortality, output, rainfall, and temperature, Bramall argues that weather played only a minor role, while central policy and local cadre responses were decisive in shaping mortality differences across Sichuan counties. This supports

the paper’s broader claim that ecological and agricultural collapse must be understood within a political system that rewarded radical mobilization and punished correction.

#### 7.4 Sichuan Sensitivity Analysis

To connect the national model to Sichuan without pretending to possess complete Sichuan-specific data, this paper introduces a mortality-output multiplier:

$$\text{Gap}_{\text{Sichuan},t} = \lambda_{\text{Sichuan}} \times \text{Gap}_{\text{National},t} \quad (8)$$

where  $\lambda_{\text{Sichuan}} = 1$  means Sichuan followed the national average, while  $\lambda_{\text{Sichuan}} > 1$  means Sichuan suffered stronger policy-famine pressure than the national mean. This is not a direct estimate of Sichuan’s actual grain output. It is a sensitivity analysis showing how the national policy-gap result would scale if Sichuan’s famine intensity were higher than average. The national model identifies the timing and

Table 4: Sichuan Multiplier Sensitivity Analysis

| $\lambda$ | Mean Gap Index | 90% Lower | 90% Upper |
|-----------|----------------|-----------|-----------|
| 1.0       | 43.3           | 28.3      | 55.9      |
| 1.5       | 65.0           | 42.4      | 83.8      |
| 2.0       | 86.7           | 56.5      | 111.7     |
| 2.5       | 108.4          | 70.6      | 139.7     |

scale of policy-related output loss. The Sichuan case then shows why some provinces may have experienced a stronger version of the national crisis. In Sichuan, the gap between official abundance and actual rural distress appears especially large.

This also strengthens the ecological argument. The *People’s Daily* reports did not merely exaggerate production; Even worse, they praised forms of mobilization that could damage the agricultural system. The October 21 report described emergency harvesting, deep plowing, labor mobilization, and militarized organization. These practices were politically celebrated as signs of revolutionary energy, but they could also disrupt normal agricultural timing, weaken local land management, and intensify pressure on soil and labor. Therefore, Sichuan illustrates the same mechanism modeled nationally: political pressure increased ecological vulnerability, and ecological vulnerability made output collapse more severe.

In short, Sichuan should be treated as a high-intensity provincial illustration of the national model. The province was represented by state media as a triumph of Maoist agricultural policy, yet it experienced extreme famine mortality soon afterward. Sadly, this contradiction is not accidental. It reveals the central mechanism of the Great Leap Forward famine: the political system converted exaggerated production claims into procurement pressure, procurement pressure reduced rural food retention, and radical mobilization weakened the ecological and labor foundations of production.

## 8 Conclusion

The results of the model support the central claim of this paper: the Great Leap Forward famine should not be understood only as a political procurement failure

or only as a weather-related food shortage. The national data show that grain output collapsed precisely when procurement pressure increased and retained grain per capita fell. The Bayesian latent-state model further suggests that the crisis years were associated with a sharp temporary rise in ecological vulnerability.

This does not prove direct soil organic matter decline in a laboratory sense. The data are not sufficient. However, the model shows that the agricultural system behaved as if its resilience had been weakened. The timing of the latent vulnerability index fits the historical evidence discussed earlier: deep plowing, close planting, labor diversion, deforestation, communal organization, and pseudoscientific campaigns all created conditions under which soil structure, water retention, and local ecological balance could deteriorate.

The case study of Sichuan province strengthens this conclusion. The *People's Daily* portrayed Sichuan in October 1958 as a grain miracle, claiming that the province had become a "thousand-jin province" and that its population would average more than 1,300 jin (650 kg) of grain per person. Yet demographic evidence shows that Sichuan soon became one of the most severely affected famine provinces. This contrast suggests that official reports did not simply describe reality. Further detrimental, they helped produce a distorted policy environment in which extraction and mobilization continued even as rural conditions deteriorated.

The paper therefore treats soil degradation not as a separate alternative explanation to political failure of Mao, but one of the mechanisms through which political failure became catastrophic. Maoist agricultural policies damaged production incentives, distorted food distribution, and weakened the ecological base of farming.

## References

- Communist Party of China. *Resolution on Certain Questions in the History of Our Party since the Founding of the People's Republic of China*. Adopted by the 6th Plenary Session of the 11th Central Committee of the CPC, June 27, 1981.
- Encyclopaedia Britannica. "Reconstruction and Consolidation, 1949–52." <https://www.britannica.com/place/China/Reconstruction-and-consolidation-1949-52>
- Encyclopaedia Britannica. "First Five-Year Plan." <https://www.britannica.com/place/China/First-Five-Year-Plan>
- Encyclopaedia Britannica. "The Transition to Socialism, 1953–57." <https://www.britannica.com/place/China/The-transition-to-socialism-1953-57>
- Mao Zedong. "Talk with the American Correspondent Anna Louise Strong," August 1946. Marxists Internet Archive. [https://www.marxists.org/reference/archive/mao/selected-works/volume-4/mswv4\\_13.htm](https://www.marxists.org/reference/archive/mao/selected-works/volume-4/mswv4_13.htm)
- Tian Zhu. *Catching Up to America: Culture, Politics, and the Emulation of Economic Success*. Cambridge: Cambridge University Press, 2023.
- Chinese Communist Party News Network. "The Second Session of the Eighth National Congress of the Chinese Communist Party Convened in Beijing on May 5, 1958."
- Song Fengying. "The Full Story of Fang Fang's Wrongful Persecution" [党史文苑]. *Party History Review* [党史文苑], no. 9 (2007): 13–17.
- People's Daily, August 27, 1958, p. 3. <https://cn.govopendata.com/renminribao/1958/8/27/3/>
- Daniel Houser, Barbara Sands, and Erte Xiao. "Three Parts Natural, Seven Parts Man-Made: Bayesian Analysis of China's Great Leap Forward Demographic Disaster." *Journal of Economic Behavior & Organization* 69, no. 2 (2009): 148–159. doi:10.1016/j.jebo.2007.09.008
- Wei Li and Dennis Tao Yang. "The Great Leap Forward: Anatomy of a Central Planning Disaster." *Journal of Political Economy* 113, no. 4 (2005): 840–877. doi:10.1086/430804
- FAO and ITPS. *Status of the World's Soil Resources: Main Report*. Rome: Food and Agriculture Organization of the United Nations and Intergovernmental Technical Panel on Soils, 2015.
- R. Lal. "Soil Carbon Sequestration Impacts on Global Climate Change and Food Security." *Science* 304, no. 5677 (2004): 1623–1627. doi:10.1126/science.1097396
- IPCC. *Climate Change and Land: An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems*. Geneva: Intergovernmental Panel on Climate Change, 2019.



Xin Meng, Nancy Qian, and Pierre Yared. “The Institutional Causes of China’s Great Famine, 1959–61.” NBER Working Paper No. 16361, 2010.

People’s Daily. “Sichuan Grain and Cotton Output More Than Doubled” [四川粮棉增产一倍多]. October 15, 1958, p. 3.

People’s Daily. “Seventy Million People Average More Than 1,300 Jin of Grain Per Person: Sichuan Becomes a Thousand-Jin Grain Province” [七千万人每人平均有粮一千三百多斤：四川跃居粮食千斤省]. October 21, 1958, p. 1.

Chris Bramall. “Agency and Famine in China’s Sichuan Province, 1958–1962.” *The China Quarterly* 208 (2011): 990–1008.

Xizhe Peng. “Demographic Consequences of the Great Leap Forward in China’s Provinces.” *Population and Development Review* 13, no. 4 (1987): 639–670.

## A Python Code for Bayesian Latent-State Production Model

The following code implements the Bayesian latent ecological-vulnerability production model described in Section 5. It uses MAP estimation via BFGS optimization and a Laplace approximation for posterior inference.

```
1  import numpy as np
2  import pandas as pd
3  from scipy.optimize import minimize
4  from scipy.special import expit, logit
5  from scipy.stats import norm, beta as beta_dist
6  rng = np.random.default_rng(20260521)
7  # -----
8  # 1. National data
9  # -----
10 data = [
11     (1952,164,33,260,173,124,76,0.3,0.08),
12     (1953,167,47,242,177,127,81,0.4,0.12),
13     (1954,170,51,228,182,129,85,0.5,0.16),
14     (1955,184,48,256,186,130,88,0.8,0.24),
15     (1956,193,40,284,185,136,88,1.1,0.33),
16     (1957,195,46,273,193,134,84,1.7,0.37),
17     (1958,200,52,268,155,128,78,2.4,0.55),
18     (1959,170,64,193,163,116,79,3.4,0.54),
19     (1960,143,47,182,170,122,73,5.0,0.66),
20     (1961,148,37,209,197,121,69,7.1,0.45),
21     (1962,160,32,229,213,122,70,10,0.63),
22     (1963,170,37,231,220,121,75,12,1.0),
23     (1964,188,40,256,228,122,79,13,1.3),
24     (1965,195,39,261,234,120,84,15,1.9),
25     (1966,214,41,282,243,121,87,17,2.3),
26     (1967,218,41,281,252,119,90,20,2.4),
27     (1968,209,40,261,261,116,92,22,2.7),
28     (1969,211,38,259,271,118,92,26,3.1),
29     (1970,240,46,282,278,119,94,29,3.4),
30     (1971,250,44,293,284,121,95,38,3.8),
31     (1972,241,39,298,283,121,96,50,4.3),
32     (1973,265,48,293,289,121,97,65,4.8),
33     (1974,275,47,303,292,121,98,81,5.4),
34     (1975,285,53,304,295,121,97,102,6.0),
35     (1976,286,49,306,294,121,95,117,6.8),
36     (1977,283,48,300,293,120,94,140,7.6),
37 ]
38 cols = ['year','grain_output','procurement','retained_pc',
39         'rural_labor','area_sown','draft_animals',
40         'machinery_hp','fertilizer']
41 df = pd.DataFrame(data, columns=cols)
42 T = len(df)
43 # -----
44 # 2. Derived policy-stress variables
45 # -----
46 df['proc_rate'] = df.procurement / df.grain_output
47 df['glf'] = ((df.year >= 1958) & (df.year <= 1961)).astype(float)
48 base_proc = df.loc[df.year <= 1957, 'proc_rate'].mean()
49 base_labor = float(df.loc[df.year == 1957, 'rural_labor'].iloc
50                    [0])
51 base_land_worker = float(
52     df.loc[df.year == 1957, 'area_sown'].iloc[0] /
```

```

52     df.loc[df.year == 1957, 'rural_labor'].iloc[0]
53 )
54 df['proc_excess'] = np.maximum(0, df.proc_rate - base_proc)
55 df['labor_diversion'] = df.glf * np.maximum(
56     0, (base_labor - df.rural_labor) / base_labor)
57 df['land_intensity'] = df.glf * np.maximum(
58     0, (df.area_sown / df.rural_labor) / base_land_worker - 1)
59 def standardize(v):
60     v = np.asarray(v, dtype=float)
61     return (v - v.mean()) / v.std()
62 stress_base = np.vstack([
63     df.glf.to_numpy(float),
64     standardize(df.proc_excess),
65     standardize(df.labor_diversion),
66     standardize(df.land_intensity)
67 ]).T
68 stress_names = ['GLF dummy', 'excess procurement',
69                 'labor diversion', 'land intensity']
70 # Production inputs: standardized logs
71 for c in ['rural_labor', 'area_sown', 'draft_animals',
72          'machinery_hp', 'fertilizer']:
73     df['log_' + c] = np.log(df[c])
74 input_cols = ['log_rural_labor', 'log_area_sown',
75              'log_draft_animals', 'log_machinery_hp',
76              'log_fertilizer']
77 X = df[input_cols].to_numpy(float)
78 Xz = (X - X.mean(axis=0)) / X.std(axis=0)
79 y_log = np.log(df.grain_output.to_numpy(float))
80 y_mean, y_sd = y_log.mean(), y_log.std()
81 yz = (y_log - y_mean) / y_sd
82 # -----
83 # 3. Bayesian latent ecological-vulnerability production model
84 # -----
85 pdim = 1 + 5 + 4 + 1 + 1 + 1 # 13 parameters
86 def unpack(theta):
87     i = 0
88     alpha = theta[i]; i += 1
89     beta = np.exp(theta[i:i+5]); i += 5
90     load = np.exp(theta[i:i+4]); i += 4
91     rho = expit(theta[i]); i += 1
92     penalty = np.exp(theta[i]); i += 1
93     sigma = np.exp(theta[i]); i += 1
94     return alpha, beta, load, rho, penalty, sigma
95 def raw_soil(load, rho, stress_matrix):
96     stress = stress_matrix @ load
97     D = np.zeros(T)
98     for t in range(T):
99         D[t] = stress[t] + (rho * D[t-1] if t > 0 else 0.0)
100     return D
101 def soil_index(load, rho, stress_matrix=stress_base,
102               reference=None):
103     D = raw_soil(load, rho, stress_matrix)
104     if reference is None:
105         reference = raw_soil(load, rho, stress_base)
106     return (D - reference.mean()) / reference.std()
107 def log_prior(theta):
108     alpha, beta, load, rho, penalty, sigma = unpack(theta)
109     lp = norm.logpdf(alpha, 0, 1)

```

```

110     lp += (np.log(2) + norm.logpdf(beta, 0, 0.45)
111            + np.log(beta)).sum()
112     lp += (np.log(2) + norm.logpdf(load, 0, 0.60)
113            + np.log(load)).sum()
114     lp += beta_dist.logpdf(rho, 6, 2) + np.log(rho) \
115            + np.log1p(-rho)
116     lp += np.log(2) + norm.logpdf(penalty, 0, 0.50) \
117            + np.log(penalty)
118     lp += np.log(2) - 2*sigma + np.log(sigma)
119     return lp
120 def log_likelihood(theta):
121     alpha, beta, load, rho, penalty, sigma = unpack(theta)
122     D = soil_index(load, rho)
123     mu = alpha + Xz @ beta - penalty * D
124     return norm.logpdf(yz, mu, sigma).sum()
125 def log_posterior(theta):
126     lp = log_prior(theta)
127     if not np.isfinite(lp):
128         return -np.inf
129     return lp + log_likelihood(theta)
130 def objective(theta):
131     val = log_posterior(theta)
132     return 1e50 if not np.isfinite(val) else -val
133 init = np.zeros(pdim)
134 init[1:6] = np.log([0.10, 0.15, 0.05, 0.15, 0.05])
135 init[6:10] = np.log([0.50, 0.80, 0.25, 0.15])
136 init[10] = logit(0.85)
137 init[11] = np.log(0.30)
138 init[12] = np.log(0.25)
139 res = minimize(objective, init, method='BFGS',
140               options={'maxiter': 20000, 'gtol': 1e-6})
141 print('Optimization success:', res.success, '|', res.message)
142 print('Negative log posterior at MAP:', round(res.fun, 3))
143 # Laplace posterior simulation
144 map_theta = res.x
145 cov = np.asarray(res.hess_inv)
146 cov = (cov + cov.T) / 2
147 vals, vecs = np.linalg.eigh(cov)
148 vals = np.clip(vals, 1e-8, 10.0)
149 cov = vecs @ np.diag(vals) @ vecs.T
150 samples = rng.multivariate_normal(map_theta, cov, size=20000)
151 # Transform posterior draws
152 parts = [unpack(th) for th in samples]
153 alpha_s = np.array([p[0] for p in parts])
154 beta_s = np.array([p[1] for p in parts])
155 load_s = np.array([p[2] for p in parts])
156 rho_s = np.array([p[3] for p in parts])
157 penalty_s = np.array([p[4] for p in parts])
158 sigma_s = np.array([p[5] for p in parts])
159 def row(name, values):
160     q = np.quantile(values, [0.05, 0.50, 0.95])
161     return {'parameter': name,
162            'mean': values.mean(),
163            'q05': q[0], 'median': q[1], 'q95': q[2]}
164 rows = [row('soil_output_penalty', penalty_s),
165         row('soil_persistence_rho', rho_s),
166         row('sigma', sigma_s)]
167 for j, name in enumerate(stress_names):

```

```

168     rows.append(row('stress_loading_' + name, load_s[:, j]))
169 for j, name in enumerate(input_cols):
170     rows.append(row('input_elasticity_' + name, beta_s[:, j]))
171 summary = pd.DataFrame(rows)
172 print('\nPosterior summary (standardized log-output units; 90%
      interval):')
173 print(summary.to_string(
174     index=False,
175     formatters={'mean': '{:.3f}'.format, 'q05': '{:.3f}'.format,
176                'median': '{:.3f}'.format, 'q95': '{:.3f}'.format}
177 ))
178 # Latent vulnerability table
179 D_samples = np.array([
180     soil_index(load_s[i], rho_s[i]) for i in range(len(samples))
181 ])
182 latent = pd.DataFrame({
183     'year': df.year,
184     'soil_index_mean': D_samples.mean(axis=0),
185     'q05': np.quantile(D_samples, 0.05, axis=0),
186     'q95': np.quantile(D_samples, 0.95, axis=0)
187 })
188 print('\nLatent ecological vulnerability index (z-score; 90%
      interval):')
189 print(latent.to_string(
190     index=False,
191     formatters={'soil_index_mean': '{:.2f}'.format,
192                'q05': '{:.2f}'.format, 'q95': '{:.2f}'.format}
193 ))
194 # Counterfactual prediction
195 def predict(theta, glf_vec, proc_excess_vec,
196             labor_div_vec, land_intensity_vec):
197     alpha, beta, load, rho, penalty, sigma = unpack(theta)
198     proc_z = (proc_excess_vec - df.proc_excess.mean()) \
199             / df.proc_excess.std()
200     labor_z = (labor_div_vec - df.labor_diversion.mean()) \
201             / df.labor_diversion.std()
202     land_z = (land_intensity_vec - df.land_intensity.mean()) \
203             / df.land_intensity.std()
204     stress_cf = np.vstack([glf_vec, proc_z, labor_z, land_z]).T
205     D = soil_index(load, rho, stress_cf,
206                   reference=raw_soil(load, rho, stress_base))
207     mu = alpha + Xz @ beta - penalty * D
208     return np.exp(mu * y_sd + y_mean)
209 obs_pred = predict(map_theta,
210                    df.glf.to_numpy(float),
211                    df.proc_excess.to_numpy(float),
212                    df.labor_diversion.to_numpy(float),
213                    df.land_intensity.to_numpy(float))
214 glf_cf = df.glf.to_numpy(float).copy()
215 glf_cf[df.glf == 1] = 0
216 proc_cf = df.proc_excess.to_numpy(float).copy()
217 proc_cf[df.glf == 1] = 0
218 labor_cf = df.labor_diversion.to_numpy(float).copy()
219 labor_cf[df.glf == 1] = 0
220 land_cf = df.land_intensity.to_numpy(float).copy()
221 land_cf[df.glf == 1] = 0
222 cf_pred = predict(map_theta,
223                    glf_cf, proc_cf, labor_cf, land_cf)

```

```

224 cf = pd.DataFrame({
225     'year': df.year,
226     'observed_output': df.grain_output,
227     'model_fit': obs_pred,
228     'no_GLF_policy_counterfactual': cf_pred,
229     'policy_gap_million_tons': cf_pred - df.grain_output
230 })
231 print('\nCounterfactual policy-removal scenario, MAP prediction:')
232 print(cf[(cf.year >= 1958) & (cf.year <= 1962)].to_string(
233     index=False,
234     formatters={
235         'model_fit': '{:.1f}'.format,
236         'no_GLF_policy_counterfactual': '{:.1f}'.format,
237         'policy_gap_million_tons': '{:.1f}'.format
238     }
239 ))
240 # Posterior interval for national gap
241 avg_gaps = []
242 for th in samples[:,20]:
243     p = predict(th, glf_cf, proc_cf, labor_cf, land_cf)
244     actual = df.loc[(df.year >= 1959) & (df.year <= 1961),
245         'grain_output'].to_numpy()
246     avg_gaps.append(
247         np.mean(p[(df.year >= 1959) & (df.year <= 1961)] - actual
248     )
249 avg_gaps = np.array(avg_gaps)
250 print(
251     '\nAverage 1959-61 national policy gap '
252     '(posterior, million tons/year): '
253     'mean={:.1f}, 90% interval [{:.1f}, {:.1f}]'.format(
254         avg_gaps.mean(), *np.quantile(avg_gaps, [0.05, 0.95])
255     )
256 )
257 # Sichuan sensitivity module
258 scenarios = []
259 for lam in [1.0, 1.5, 2.0, 2.5]:
260     vals = avg_gaps * lam
261     scenarios.append({
262         'lambda': lam,
263         'mean_gap_index': vals.mean(),
264         'q05': np.quantile(vals, 0.05),
265         'q95': np.quantile(vals, 0.95)
266     })
267 print('\nSichuan multiplier sensitivity:')
268 print(pd.DataFrame(scenarios).to_string(
269     index=False,
270     formatters={
271         'lambda': '{:.1f}'.format,
272         'mean_gap_index': '{:.1f}'.format,
273         'q05': '{:.1f}'.format,
274         'q95': '{:.1f}'.format
275     }
276 ))

```